

SCENARIO BASED ACTIVITY.

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COURSE : COMPUTER
VISION.

Mention the techniques used with reason :

1. A CCTV image contains Salt-and-pepper noise due to transmission errors.

Median Filter - Removes Salt-and-pepper noise effectively while preserving edges.

2. A Satellite image is corrupted by Gaussian noise from sensors

Gaussian Filter - Reduces Gaussian noise by smoothing the image.

3. A medical X-ray image has random noise but fine edges must be preserved.

Bilateral Filter - Removes random noise while preserving fine edges.

4. A scanned document has black and white dots scattered randomly.

Median Filter - Eliminates scattered black-and-white dots in scanned documents.

5. A low-light image shows heavy grain without clear boundaries.

Non-local Means [NLM] Filter - Reduces heavy grain / noise while preserving image details.

6. A foggy road image has very low contrast.

Histogram Equalization - Enhances the contrast of the foggy image.

7. An underwater image appears dull and washed out.

Contrast Stretching - Improves brightness and contrast in dull underwater images.

8. A chest x-ray shows poor visibility of internal structures.

CLAHE [Contrast Limited Adaptive Histogram Equalization] - Enhances local contrast and improves visibility of internal structures in chest x-ray.

9. A night-time traffic image lacks intensity variation.

Histogram Equalization - Increases intensity variation in night-time traffic images.

10. A grayscale image uses only a narrow intensity range.

Contrast Stretching - Expands the narrow grayscale intensity range to improve image visibility.